

## Chapter 2, Section 9

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1. Let  $X$  be a Noetherian scheme,  $Y$  a closed subscheme, and  $\hat{X}$  the completion of  $X$  along  $Y$ . We call the ring  $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}})$  the ring of *formal-regular functions* on  $X$  along  $Y$ . In this exercise we show that if  $Y$  is a connected, nonsingular, positive dimensional subvariety of  $X = \mathbb{P}_k^n$  over an algebraically closed field  $k$ , then  $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}}) = k$ .

(a) Let  $\mathcal{I}$  be the ideal sheaf of  $Y$ . Use (8.13) and (8.17) to show that here is an inclusion of sheaves on  $Y, \mathcal{I}/\mathcal{I}^2 \hookrightarrow \mathcal{O}_Y(-1)^{n+1}$ .

(b) Show that for any  $r \geq 1$ ,  $\Gamma(Y, \mathcal{I}^r/\mathcal{I}^{r+1}) = 0$ .

(c) Use the exact sequences

$$0 \rightarrow \mathcal{I}^r/\mathcal{I}^{r+1} \rightarrow \mathcal{O}_X/\mathcal{I}^{r+1} \rightarrow \mathcal{O}_X/\mathcal{I}^r \rightarrow 0$$

and induction on  $r$  to show that  $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$  for all  $r \geq 1$ .

(d) Conclude that  $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}}) = k$ .

2. Use the result of (Ex. 9.1) to prove the following geometric result. Let  $Y \subseteq X = \mathbb{P}_k^n$  be as above, and let  $f : X \rightarrow Z$  be a morphism of  $k$ -varieties. Suppose that  $f(Y)$  is a single closed point  $P \in Z$ . Then  $f(X) = P$  also.

3. Prove the analogue of (5.6) for formal schemes, which says, if  $\mathfrak{X}$  is an affine formal scheme, and if

$$0 \rightarrow \mathfrak{F}' \rightarrow \mathfrak{F} \rightarrow \mathfrak{F}'' \rightarrow 0$$

is an exact sequence of  $\mathcal{O}_{\mathfrak{X}}$ -modules, and if  $\mathfrak{F}'$  is coherent, then the sequence of global sections

$$0 \rightarrow \Gamma(\mathfrak{X}, \mathfrak{F}') \rightarrow \Gamma(\mathfrak{X}, \mathfrak{F}) \rightarrow \Gamma(\mathfrak{X}, \mathfrak{F}'') \rightarrow 0$$

is exact. For the proof, proceed in the following steps.

(a) Let  $\mathfrak{J}$  be an ideal of definition for  $\mathfrak{X}$ , and for each  $n > 0$  consider the exact sequence

$$0 \rightarrow \mathfrak{F}'/\mathfrak{J}^n \mathfrak{F}' \rightarrow \mathfrak{F}/\mathfrak{J}^n \mathfrak{F} \rightarrow \mathfrak{F}''/\mathfrak{J}^n \mathfrak{F}'' \rightarrow 0.$$

Use (5.6), slightly modified, to show that for every open affine subset  $\mathfrak{U} \subseteq \mathfrak{X}$ , the sequence

$$0 \rightarrow \Gamma(\mathfrak{U}, \mathfrak{F}'/\mathfrak{J}^n \mathfrak{F}') \rightarrow \Gamma(\mathfrak{U}, \mathfrak{F}/\mathfrak{J}^n \mathfrak{F}) \rightarrow \Gamma(\mathfrak{U}, \mathfrak{F}'') \rightarrow 0$$

is exact.

(b) Now pass to the limit, using (9.1), (9.2), and (9.6). Conclude that  $\mathfrak{F} \cong \varprojlim \mathfrak{F}/\mathfrak{J}^n \mathfrak{F}$  and that the sequence of global sections above is exact.

4. Use (Ex. 9.3) to prove that if

$$0 \rightarrow \mathfrak{F}' \rightarrow \mathfrak{F} \rightarrow \mathfrak{F}'' \rightarrow 0$$

is an exact sequence of  $\mathcal{O}_{\mathfrak{X}}$ -modules on a Noetherian formal scheme  $\mathfrak{X}$ , and if  $\mathfrak{F}', \mathfrak{F}''$  are coherent, then  $\mathfrak{F}$  is coherent.

5. If  $\mathcal{F}$  is a coherent sheaf on a Noetherian formal scheme  $\mathfrak{X}$ , which can be generated by global sections, show in fact that it can be generated by a finite number of its global sections.

6. Let  $\mathfrak{X}$  be a Noetherian formal scheme, let  $\mathfrak{J}$  be an ideal of definition, and for each  $n$ , let  $Y_n$  be the scheme  $(\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathfrak{J}^n)$ . Assume that the inverse system of groups  $(\Gamma(Y_n, \mathcal{O}_{Y_n}))$  satisfies the Mittag-Leffler condition. Then prove that  $\text{Pic } \mathfrak{X} = \varprojlim \text{Pic } Y_n$ . As in the case of a scheme, we define  $\text{Pic } \mathfrak{X}$  to be the group of locally free  $\mathcal{O}_{\mathfrak{X}}$ -modules of rank 1 under the operation  $\otimes$ . Proceed in the following steps.

- (a) Use the fact that  $\ker(\Gamma(Y_{n+1}, \mathcal{O}_{Y_{n+1}}) \rightarrow \Gamma(Y_n, \mathcal{O}_{Y_n}))$  is a nilpotent ideal to show that the inverse system  $(\Gamma(Y_n, \mathcal{O}_{Y_n}^*))$  of units in the respective rings also satisfies (ML).
- (b) Let  $\mathfrak{F}$  be a coherent sheaf of  $\mathcal{O}_{\mathfrak{X}}$ -modules, and assume that for each  $n$ , there is some isomorphism  $\varphi_n : \mathfrak{F}/\mathfrak{I}^n \mathfrak{F} \cong \mathcal{O}_{Y_n}$ . Be careful, because the  $\varphi_n$  may not be compatible with the maps in the two inverse systems  $(\mathfrak{F}/\mathfrak{I}^n \mathfrak{F})$  and  $(\mathcal{O}_{Y_n})$ ! Then show that there is an isomorphism  $\mathfrak{F} \cong \mathcal{O}_{\mathfrak{X}}$ . Conclude that the natural map  $\text{Pic } \mathfrak{X} \rightarrow \varprojlim \text{Pic } Y_n$  is injective.
- (c) Given an invertible sheaf  $\mathcal{L}_n$  on  $Y_n$  for each  $n$ , and given isomorphism  $\mathcal{L}_{n+1} \otimes \mathcal{O}_{Y_n} \cong \mathcal{L}_n$ , construct maps  $\mathcal{L}_{n'} \rightarrow \mathcal{L}_n$  for each  $n' \geq n$  so as to make an inverse system, and show that  $\mathfrak{L} = \varprojlim \mathcal{L}_n$  is a coherent sheaf on  $\mathfrak{X}$ . Then show that  $\mathfrak{L}$  is locally free of rank 1, and thus conclude that the map  $\text{Pic } \mathfrak{X} \rightarrow \varprojlim \text{Pic } Y_n$  is surjective. Again be careful, because even though each  $\mathcal{L}_n$  is locally free of rank 1, the open sets needed to make them free might get smaller and smaller with  $n$ .
- (d) Show that the hypothesis “ $(\Gamma(Y_n, \mathcal{O}_{Y_n}))$  satisfies (ML)” is satisfied if either  $\mathfrak{X}$  is affine, or each  $Y_n$  is projective over a field  $k$ .